Dianchao Wu

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Education

Iowa State University

B.S. Mathematics — Expected May 2027

Cumulative GPA: 3.85/4.00 · **Mathematics GPA:** 3.89/4.00

Research Experience

Degenerate Diffusions in Finite and Infinite Dimensions: Smoothing and Convergence

Student Researcher · Advisor: David P. Herzog — Summer 2026-Present

NSF-funded research · Award DMS-2246491

- Formulated the Berry-Esseen framework for the unwrapped displacement and proved the overdamped results.
- Proved the main underdamped theorem and supporting propositions via a direct Markov-additive treatment of the displacement; verification of Assumption (A2) was carried out by O. Smith.
- Proved in Proposition 11.1 that, for one-dimensional periodic underdamped dynamics started from equilibrium and sufficiently small positive γ, the Berry-Esseen constant C_γ can be chosen with $C_\gamma \le C\gamma^{-1/2}$, where C is independent of γ.
- Carried out the original numerical illustration and drafted Part I.

Numerical study of the low-friction regime: A. Valse.

Nonlinear MBS Pricing Models

Student Researcher · Advisor: Rana D. Parshad — 2026-Present

- Study whether stochastic perturbations and nonlinear p-Laplacian diffusion improve the predictive performance of an MBS pricing equation.
- Built an initial backtesting workflow using real transaction data and examined numerical behavior and financial interpretation; a stable predictive improvement has not yet been established.
- Next stage: stochastic analysis, mean-square sensitivity, noise-driven simulation, and model-risk interpretation.

Random Matrix Theory and Quantum Expanders

ISmART Research Project · Mentor: Dr. Jananan Arulseelan — Spring-Summer 2026

Past exploratory study of Haar-distributed unitary matrices and quantum expanders.

Manuscripts

Berry-Esseen Bounds for Periodic Langevin Dynamics

Manuscript in preparation

O. Smith, A. Valse, D. Wu (alphabetical). Mentor: Professor David P. Herzog.

Prediction, Model Risk, and Stochastic Sensitivity in Nonlinear MBS Pricing Models with *p*-Laplacian Diffusion

In Progress

Directed Studies

Four faculty-supervised reading courses.

Stochastic Calculus | David P. Herzog

Text: Bernt Øksendal, *Stochastic Differential Equations: An Introduction with Applications*, 6th ed.

Brownian motion, Itô integration and formula, SDEs, diffusion generators, and stochastic dynamics.

Foundations of Probability | Ruoyu Wu

Text: Ross and Leadbetter, *A Basic Course in Measure and Probability*.

Measure-theoretic probability, random variables, integration, independence, and modes of convergence.

Real Analysis and Measure Theory | Ruoyu Wu

Text: H. L. Royden and P. M. Fitzpatrick, *Real Analysis*, 4th ed.

Lebesgue measure and integration, convergence theorems, and L^p spaces.

Partial Differential Equations I | Rana D. Parshad

Text: Lawrence C. Evans, *Partial Differential Equations*.

Sobolev spaces, weak derivatives and solutions, and functional-analytic methods for PDE.

Selected Coursework

In progress · Fall 2026

MATH 5010 - Introduction to Real Analysis; **MATH 5540** - Stochastic Process Models.

Planned · Spring 2027

MATH/STAT 6450 - Advanced Stochastic Processes; **MATH 5250** - Numerical Analysis of High-Performance Computing; **STAT 5279** - Statistical Computing for Data Analysis.

Selected coursework · Completed

MATH 3850 · Introduction to Partial Differential Equations; MATH 4030 · Intermediate Abstract Algebra; MATH 4430X · Introduction to Mathematical Finance; MATH 3010 · Abstract Algebra I; MATH 3170 · Theory of Linear Algebra; MATH 4900 · Independent Study, 6 credits.

Honors & Awards

Fred Wright Mathematics Endowed Scholarship	May 2026
ISU Athletics O.S.C.A.R.S. Award	May 2026
Big 12 Conference All-Academic Team	Spring 2026
CSC Academic All-District	2025
Academic All-Big 12 First Team	2025
Academic All-Big 12 Rookie Team	2024
Big 12 Commissioner's Honor Roll	Fall 2023-Fall 2025

Athletics

Iowa State University Men's Golf

NCAA Division I Student-Athlete — August 2023-Present

Represent Iowa State University in NCAA Division I tournaments. Balance more than 20 hours of weekly training and competition with a full mathematics course load. Best World Amateur Golf Ranking: 443.

Technical Skills

Programming: Python, MATLAB, C (currently learning)

Scientific work: Numerical simulation, stochastic-process simulation, market-data analysis

Typesetting: LaTeX

Languages: English (fluent); Mandarin and Hakka (native)